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Starting the proof of

toric mirror Symmetry

What we've done so far

- Background on toric geometry
- Whirlwind tour of ∞-categories
- (Commutative) monoids & spectra
- The functor of points
- Set up basic spectral algebraic geometry
  - We can then immediately formulate toric geometry over the universal base, the sphere spectrum

Rest of the semester Proving toric mirror symmetry. Let's recall the most basic statement.

Ntn Given a fan (N,Σ), we write $X_\sigma = \mathrm{Spec}(\mathbb{S}\langle M \cap \sigma^\vee \rangle)$ and $X_\Sigma = \operatorname{Colim}_{\sigma \in \Sigma} X_\sigma$ for the toric stack over the sphere.

- Nothing specific about the base will be used, so if it makes it mentally easier, feel free to replace $\mathbb{S}$ by $\mathbb{Z}$ or a field k
  - And $\mathrm{Sp} = \mathrm{Mod}_{\mathbb{S}}$ by $\mathrm{Mod}_{\mathbb{Z}}$ or Mod_k.

Recall. A fan (N,Σ) is:

(1) **Smooth** if each cone $\sigma \in \Sigma$ can be generated by part of a basis for N

(2) **projective** if there is a rational polytope $P \subset M_{\mathbb{R}}$ such that of full dimension such that Σ is the normal fan Σ_P of P

Thm (FLTZ) Let (N,Σ) be a smooth projective fan. There is a fully faithful symmetric monoidal functor

Convolution

$$\kappa : (QCoh(X_\Sigma / T), \otimes) \hookrightarrow (Sh(M_{\mathbb{R}}; Sp), *)$$

The image of κ consists of those sheaves with **microsupport** in the $\mathbb{R}_{>0}$-conic Lagrangian Λ_Σ in

$$T^* M_{\mathbb{R}} \simeq M_{\mathbb{R}} \times N_{\mathbb{R}}$$

defined by

$$\Lambda_\Sigma := \bigcup_{m \in M, \sigma \in \Sigma} (m + \sigma^\vee) \times \sigma.$$

Proof Overview

Strategy to construct κ.

(1) We start with an affine version of the problem. For $\sigma \in \Sigma$, we want a combinatorial description of $QCoh(X_\sigma/T)$.

- We'll define a poset $\Theta(\sigma)$ of some specific closed polyhedral subsets of $M_{\mathbb{R}}$. This will be a symmetric monoidal category under **Minkowski Sum**

$$S+T := \{s+t \mid s\in S \text{ and } t\in T\}.$$

- We'll construct a symmetric monoidal equivalence

$$(\mathrm{Fun}(\Theta(\sigma)^{op}, Sp), \otimes_{Day}) \xrightarrow{\sim} (QCoh(X_\sigma/T), \otimes)$$

- We'll construct a fully faithful lax sym. monoidal structure

$$(\mathrm{Fun}(\Theta(\sigma)^{op}, Sp), \otimes_{Day}) \hookrightarrow (Sh(M_{\mathbb{R}}; Sp), *).$$

(2) We'll show that all the functors in (1) are compatible with inclusions of cones

- By Zariski descent, taking the limit over $\sigma \in \Sigma^{op}$, we obtain the functor K:

$$\mathrm{QCoh}(X_\Sigma/T) \simeq \lim_{\sigma \in \Sigma^{op}} \mathrm{QCoh}(X_\sigma/T) \hookrightarrow \mathrm{Sh}(M_{\mathbb{R}}, \mathrm{Sp})$$

- This will involve an analagous gluing formula for $\mathrm{Sh}(M_{\mathbb{R}}, \mathrm{Sp})$ in terms of subcategories of modules over **idempotent algebras**

Note. To construct these functors, we'll use the six-functor formalism on topological spaces. For example, when $N = \mathbb{Z}$ and $\sigma = \mathbb{R}_{>0}$, we'll have

> $\Theta(\sigma) = (\mathbb{Z}, \leq)^{op}$ with sym. mon structure addition

> The functor $\mathrm{Fun}(\mathbb{Z}, \mathrm{Sp}) \longrightarrow \mathrm{Sh}(\mathbb{R}; \mathrm{Sp})$ will send $n \in \mathbb{Z}$ to the dualizing sheaf $\omega_{(-\infty, n)} = p^!(\mathbb{S})$ $p: (-\infty, n) \longrightarrow *$

> The equivalence $\mathrm{Fun}(\mathbb{Z}, \mathrm{Sp}) \xrightarrow{\sim} \mathrm{QCoh}(\mathbb{A}^1/\mathbb{G}_m)$ is Simpson's

equivalence that classifies the universal line bundle $\mathcal{O}(1)$ with a section $\cdot x : \mathcal{O} \longrightarrow \mathcal{O}(1)$

Strategy to characterize the image of κ

> The idea is that the relationship to the combinatorial description of quasicoherent sheaves comes from the theory of **exit-path ∞-categories**:

- If (X,P) is any reasonable stratified space, there is an ∞-category $\mathrm{Exit}(X,P)$ and an equivalence

$$\mathrm{Fun}(\mathrm{Exit}(X,P), \underset{\text{or } \mathrm{Sp}}{\mathrm{An}}) \simeq \left(\begin{array}{l} \text{Sheaves on } X \text{ that} \\ \text{are locally constant} \\ \text{along the stratification} \end{array} \right)$$

'constructible sheaves'

- Moreover, this ∞-category is often a 1-category.

> The theory of sheaves with specified microsupport is a generalization of the theory of constructible sheaves, but in the context of manifolds.

- Developed by Kashiwara-Schapira
- Input: (real analytic) manifold X, $\Lambda \subset T^*X$ closed, $\mathbb{R}_{\geq 0}$-conic (subanalytic) Lagrangian
- Output: A full subcategory $Sh_\Lambda(X;Sp) \subset Sh(X;Sp)$ closed under limits and colimits. When Λ is the union of conormals to a Whitney stratification, $Sh_\Lambda(X;Sp)$ recovers constructible sheaves
- The theory is involved to set up in general, but greatly simplifies when X is a vector space
- Careful analysis is needed to identify $\mathrm{im}(k)$.

Conjecture (Nadler, vague formulation). The ∞-category $Sh_\Lambda(X;Sp)$ should admit an 'exit-path' description. That is, there should be a geometrically-defined ∞-category $\mu\mathrm{Exit}(X,\Lambda)$ and an equiv.

$$Sh_\Lambda(X;Sp) \simeq \mathrm{Fun}^{\text{conditions}}(\mu\mathrm{Exit}(X,\Lambda), Sp).$$

> Peter is very interested in this conjecture!

The combinatorial model

Def. Let N be a lattice. A subset $P \subset M_{\mathbb{R}}$ is **polyhedral** if P can be written as the Minkowski sum of a polytope and a polyhedral cone. We write $\mathrm{Poly}(M)$ for the poset of polyhedral subsets of $M_{\mathbb{R}}$ ordered by inclusion.

Observations.

(1) Every polyhedral subset is closed.

(2) The Minkowski sum of two polytopes is a polytope and the sum of two polyhedral cones is a polyhedral cone. So $\mathrm{Poly}(M)$ is closed under Minkowski sum.

(3) Minkowski sum makes $\mathrm{Poly}(M)$ into a symmetric monoidal category with unit $0 = \{0\}$.

(4) If $P \subset M_{\mathbb{R}}$ is a polyhedral cone, then $P + P = P$.

Def Let N be a lattice and $\sigma \subset N_{\mathbb{R}}$ a cone. Write

$$\Theta(\sigma) \subset \mathrm{Poly}(M)$$

for the full subposet

$$\Theta(\sigma) = \{m + \sigma^\vee \mid m \in M\}$$

Note Each $\Theta(\sigma)$ is closed under Minkowski sum, but does not usually contain the unit 0. Since $\sigma^\vee \subset M_{\mathbb{R}}$ is a cone, $\sigma^\vee + \sigma^\vee = \sigma^\vee$. So we see that $\Theta(\sigma)$ is a symmetric monoidal category under Minkowski sum, but with unit $\sigma^\vee$.

Def Let N be a lattice and $i: \sigma \hookrightarrow \tau$ an inclusion of cones in $N_{\mathbb{R}}$. Write $\Theta(i): \Theta(\tau) \longrightarrow \Theta(\sigma)$ for the functor given by

$$m + \tau^\vee \longmapsto m + \sigma^\vee.$$

Note that this functor is symmetric monoidal (since everything is a poset, there are no coherences to check, and this is a property).

The assignment $\sigma \mapsto \Theta(\sigma)$ assembles into a functor

$$\Sigma^{op} \longrightarrow \mathrm{CMon}(\mathrm{Cat}_\infty)$$

Digression: idempotent algebras

- We don't have enough time in the remainder of this lecture to construct the functor
$$\mathrm{Fun}(\Theta(\sigma)^{op}, \mathrm{Sp}) \longrightarrow \mathrm{QCoh}(X_\sigma/T)$$
and prove it is an equivalence. So, instead, we'll explain a concept that will appear lated and is related to the definition of $\Theta(\sigma)$.

Def Let $\mathcal{C}$ be a symmetric monoidal ∞-category. A map $u: \mathbb{1} \to X$ exhibits X as an **idempotent object** if
$$\mathrm{id}_X \otimes u : X \simeq X \otimes \mathbb{1} \longrightarrow X \otimes X$$
is an equivalence.
A commutative algebra $A \in \mathrm{CAlg}(\mathcal{C})$ is an **idempotent algebra** if the multiplication

- These are actually the same notion:

Thm. Let $\mathcal{C}$ be a symmetric monoidal ∞-category.

(1) If $u: \mathbb{1} \longrightarrow X$ is an idempotent object, there is a unique commutative algebra structure on X with unit u.

(2) If A is an idempotent algebra, then the unit $\mathbb{1} \longrightarrow A$ makes A into an idempotent object.

(3) The functor

$$\begin{aligned} \mathrm{CAlg}(\mathcal{C}) &\longrightarrow \mathcal{C}_{\mathbb{1}/} \\ A &\longmapsto [\text{unit } \mathbb{1} \longrightarrow A] \end{aligned}$$

restricts to an equivalence between idempotent algebras and idempotent objects.

very important

(4) If A is an idempotent algebra, then the forgetful functor $\mathrm{Mod}_A(\mathcal{C}) \longrightarrow \mathcal{C}$ is fully faithful with image those objects $Y \in \mathcal{C}$ such that $\mathrm{id}_Y \otimes \mathrm{unit}: Y \simeq Y \otimes \mathbb{1} \longrightarrow Y \otimes A$ is an equiv.

Note Item (4) says it is a **property** to be an A-module, not extra **structure**.

Ex If N is a lattice and $\sigma \subset N_{\mathbb{R}}$ is a cone, then $\sigma^\vee$ is an idempotent algebra in $\mathrm{Poly}(M)$. The poset $\Theta(\sigma)$ is a full subcategory of

$$\mathrm{Mod}_{\sigma^\vee}(\mathrm{Poly}(M))$$

Ex Let A be an ordinary ring and $S \subset A$ a subset. Then the localization is an idempotent algebra in $\mathrm{Mod}_A^\heartsuit$

Ex For any prime p, the field $\mathbb{F}_p$ is an idempotent algebra in $\mathrm{Mod}_{\mathbb{Z}}^\heartsuit$

> However, $\mathbb{F}_p$ is **not** an idempotent algebra in $\mathrm{Mod}_{\mathbb{Z}} \simeq D(\mathbb{Z})$. This is because $\mathbb{F}_p \otimes^L_{\mathbb{Z}} \mathbb{F}_p \not\simeq \mathbb{F}_p$. Instead,

$$H_1(\mathbb{F}_p \otimes^L_{\mathbb{Z}} \mathbb{F}_p) \simeq \mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{F}_p, \mathbb{F}_p) \simeq \mathbb{F}_p.$$

Ex. One can show that $\mathbb{Q}$ is an idempotent algebra in Sp, but $\mathbb{F}_p$ is not.

Idea for why $\mathbb{F}_p$ isn't an idempotent algebra In Sp and $D(\mathbb{Z})$, one needs to remember **how** $p=0$, not just **that** $p=0$.

Prop. Let $\mathcal{C}$ be a sym mon ∞-category. A commutative algebra $A \in \mathrm{CAlg}(\mathcal{C})$ is idempotent if and only if for all $B \in \mathrm{CAlg}(\mathcal{C})$, the anima $\mathrm{Map}_{\mathrm{CAlg}(\mathcal{C})}(A,B)$ is empty or terminal. Moreover, $\mathrm{Map}_{\mathrm{CAlg}(\mathcal{C})}(A,B) \neq \emptyset$ if and only if B is an A-module.

Exercise If $A, B \in \mathrm{CAlg}(\mathcal{C})$ are idempotent algebras, then $A \otimes B$ is an idempotent algebra.

Upshot For any sym mon ∞-category $\mathcal{C}$, the ∞-category $\mathrm{CAlg}^{\mathrm{idem}}(\mathcal{C})$ of idempotent algebras is a poset with binary meets (given by $\otimes$).

Smashing localizations

Def A symmetric monoidal functor $L: \mathcal{C} \to \mathcal{D}$ is a **smashing localization** if there is an idempotent algebra $A \in \mathrm{CAlg}(\mathcal{C})$ such that L factors as

$$\mathcal{C} \xrightarrow{A \otimes (-)} \mathrm{Mod}_A(\mathcal{C}) \xrightarrow{\sim} \mathcal{D}$$

> So L admits a lax sym. monoidal right adjoint F
> In this case, A is the unique commutative algebra structure on the idempotent object $F(\mathbb{1}_{\mathcal{D}}) \simeq FL(\mathbb{1}_{\mathcal{C}})$ with unit the adjunction unit $\mathbb{1}_{\mathcal{C}} \longrightarrow FL(\mathbb{1}_{\mathcal{C}})$
  - This agrees with the commutative algebra structure on $F(\mathbb{1}_{\mathcal{D}})$ coming from the fact that F is lax sym. monoidal.

Def. Let $L: \mathcal{C} \to \mathcal{D}$ be a symmetric monoidal left adjoint with right adjoint $F: \mathcal{D} \to \mathcal{C}$, and let $X \in \mathcal{C}$ and $Y \in \mathcal{D}$. Write

$$\alpha_{X,Y}: X \otimes_{\mathcal{C}} F(Y) \longrightarrow F(L(X) \otimes_{\mathcal{D}} Y)$$

for the map adjoint to

$$L(X \otimes_{\mathcal{C}} F(Y)) \simeq L(X) \otimes_{\mathcal{D}} LF(Y) \xrightarrow{\mathrm{id} \otimes \mathrm{counit}} L(X) \otimes_{\mathcal{D}} Y.$$

We say that the adjunction $L \dashv F$ **satisfies the projection formula** if for all $X \in \mathcal{C}$ and $Y \in \mathcal{D}$, the map $\alpha_{X,Y}$ is an equiv.

Recognition Principle for smashing localizations. A sym mon left adjoint $L: \mathcal{C} \longrightarrow \mathcal{D}$ is a smashing localization if and only if L admits a fully faithful right adjoint F and the adjunction $L \dashv F$ satisfies the projection formula.